

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Physiotherapy (B.P.T)

Semester – II Examination – JUNE 2012

PT108 FUNDAMENTALS OF BIOMECHANICS

Date: 5.06.12, Tuesday

Time: 10.00a.m to 10.20a.m

Maximum Marks: 10

Instructions:

1. There are two parts in the paper. **PART –I** contains 10 Questions and **PART – II** contains 4 Questions.
2. **PART –I** should be answered in MCQ 'S question paper & **PART – II** should be answered in two separate answer books which has two sections.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for **PART – I** is 20 min. Question paper for **PART – II** will be given at the table when the answer sheet for **PART – I** is returned. However, Candidates who complete **PART – I** earlier than 20 min, can collect main question paper for **PART –II** immediately after handing over the **PART-I** question paper

PART-I

Multiple Choice Questions: (MCQs)

(10 × 1 = 10)

(Encircle the correct answer)

1. The ability of any force to cause rotation of the lever is known as.....
 - a. Compression
 - b. Tension
 - c. Muscle pull
 - d. Torque
2. In case of first order levers, the relationship of the weight & effort to the fulcrum is expressed as.....
 - a. F-----E-----W
 - b. F---E-----W
 - c. F-----W-----E
 - d. W-----F-----E
3. All of the following muscles belong to thoraco-pelvic group of trunk flexors EXCEPT.....
 - a. Rectus Abdominis
 - b. Iliacus
 - c. Internal Oblique
 - d. External Oblique
4. Identify the incorrect statement pertaining to Levers.
 - a. Mechanical Advantage (M.A.) is the ratio of the weight to the effort
 - b. M.A. is obtained in levers of 1st order when the fulcrum is nearer to the weight than to the effort
 - c. M.A. is obtained in all levers of 2nd order
 - d. M.A. is obtained in all levers of 3rd order
5. Which of the following is an example of close chain activity?
 - a. Elbow Flexion
 - b. Straight Leg Raise
 - c. Stand to Sit
 - d. None of the above

6. The movement of joint surfaces is referred to as.....
- Osteokinematics
 - Arthrokinematics
 - Kinetics
 - Pathomechanics
7. Bucket handle motion causes
- AP movement of ribcage
 - Lateral motion of ribcage
 - Rotation of ribcage
 - Vertical movement of ribcage
8. Rolling motion is described as
- Motion in pure translator direction
 - Motion in which equidistant points touch each other in the course of motion
 - Motion wherein the bodies describe circular motion against each other
 - None of the above
9. Inversion- Eversion takes place at
- Talocrural joint
 - Subtalar joint
 - Talonavicular joint
 - Tarsometatarsal joint
10. The most mobile joint of the cervical spine is the articulation between.....
- C5-6
 - C6-7
 - Atlas-occipital condyles
 - Atlas-axis

CHAROTAR UNIVERSITY OF SCI. & TECHNOLOGY	
CHARUSAT CAMPUS CHANGA	
Examination Semester _____	
Subject _____ No. _____	
Candidate's Seat No. _____	
Section _____	
Date _____	
Sr. Supervisor's Sign.	

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Physiotherapy (B.P.T)

Semester – II Examination – JUNE 2012

PT108 FUNDAMENTALS OF BIOMECHANICS

Date: 5.06.12, Tuesday

Time: 10.20a.m to 1.00p.m

Maximum Marks: 60

Instructions:

1. There are two parts in the paper. PART –I contains 10 Questions and PART – II contains 4 Questions.
2. PART –I should be answered in MCQ 'S question paper & PART – II should be answered in two separate answer books which has two sections.
3. Marks on the right side suggest the total marks of that question.
4. Maximum time allotted for PART – II is 2 Hr 40min.
5. Draw the figure wherever necessary.

PART-II SECTION –I

Q.1. Long Question (1 out of 2)

(1× 10= 10)

What is force? How do you calculate force? Explain force exerted by biceps brachii at different angle of elbow flexion.

OR

What is posture? How is the plumb line passing in the sagittal plane in standing posture? What are the deviations from optimal alignment in sagittal plane?

Q.2. Write short notes on: (4 out of 5)

(4× 5 = 20)

- a. Explain types of motion with examples.
- b. Active and passive insufficiency.
- c. Stress and strain.
- d. Classification of muscles.
- e. Reflex arc.

SECTION –II

Q.3. Long Question (1 out of 2)

(1× 10= 10)

Describe the articulating surface of the Temporomandibular (TM) Joint. Describe the motions in the upper and lower joints during mouth opening and closing. Which are the signs and symptoms of TM Joint dysfunction?

OR

Write in detail about mechanics of breathing. Describe Lung volumes and capacities of Normal, Restrictive Diseases and Obstructive Diseases.

Q.4. Write short notes on: (4 out of 5)

(4× 5 = 20)

- a. Muscles of the pelvic floor.
- b. Intervertebral Discs.
- c. Axes & Planes.
- d. Fixed support synergies.
- e. Effects of Immobilization on ligament, tendon, bone and articular surfaces.